An operating system serves as an intermediary intermediary between computer hardware and user applications, managing system hardware resources efficiently. A process is an instance of a computer program in execution, consisting of program code, program counter, registers, call stack, and heap memory.

CPU scheduling determines which ready process is allocated CPU execution time. Preemptive scheduling allows the operating system to interrupt a currently executing process to assign the CPU to a higher-priority task. Round Robin (RR) is a preemptive scheduling algorithm that assigns each process a fixed time slice known as a time quantum. First-Come First-Served (FCFS) is a non-preemptive scheduling policy that can cause the convoy effect, where short processes wait behind long CPU-bound tasks. Shortest Job First (SJF) minimizes average waiting time but requires prior knowledge of process execution bursts.

Process synchronization coordinates concurrent execution to prevent race conditions when multiple processes access shared memory locations. A critical section refers to code segments that access shared resources. Mutual exclusion ensures that only one process executes within its critical section at any given moment. Semaphores and mutex locks are synchronization primitives designed to enforce mutual exclusion.

A deadlock occurs when a set of concurrent processes are permanently blocked because each process holds a resource while waiting for another resource held by another process. Four Coffman conditions must hold simultaneously for a deadlock to arise: Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait. The Banker's Algorithm is a classic deadlock avoidance strategy that checks system state safety before granting resource requests.